

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_11apr25_isolado.parquet | Experiment: rssi_odom_dataset_12apr25_isolado.parquet

model_type: LSTM
group_sec: 80
timesteps_orig: 90
batch_size: 256
min_units: 16
max_units: 64
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['elu']
optimizers: ['rmsprop']
min_layers: 1
max_layers: 1
max_epochs: 50
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 3
forecast_steps: 36
timesteps: 90

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=90
Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=90

Tuning:

Trials executed: 4/10

Training Performance:

Loss: 0.0037 | MAE: 0.0465

Validation Performance:

Loss: 0.0083 | MAE: 0.0731

LSTM Multi-Step Training Report

Report Generated: 2025-08-06 09:24:05

Training Start: 2025-08-06_09-13-43

Training End: 2025-08-06_09-15-14

Training Duration: 1.53 minutes

Predictions Start: 2025-08-06_09-15-14

Predictions End: 2025-08-06_09-19-28

Predictions Duration: 4.23 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 3.2045

MSE: 15.8535

RMSE: 3.9816

R2 Score: 0.5268

Multi-Step Forecast:

Forecast Steps: 36

MAE: 0.1087

MSE: 0.0177

RMSE: 0.1331

R2 Score: 0.6052

Dataset size before down-sample: 31,097 rows
Dataset size after down-sample: 4,672 rows

First 5 timestamps BEFORE:

1744464597710
1744464597720
1744464597744
1744464597757
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

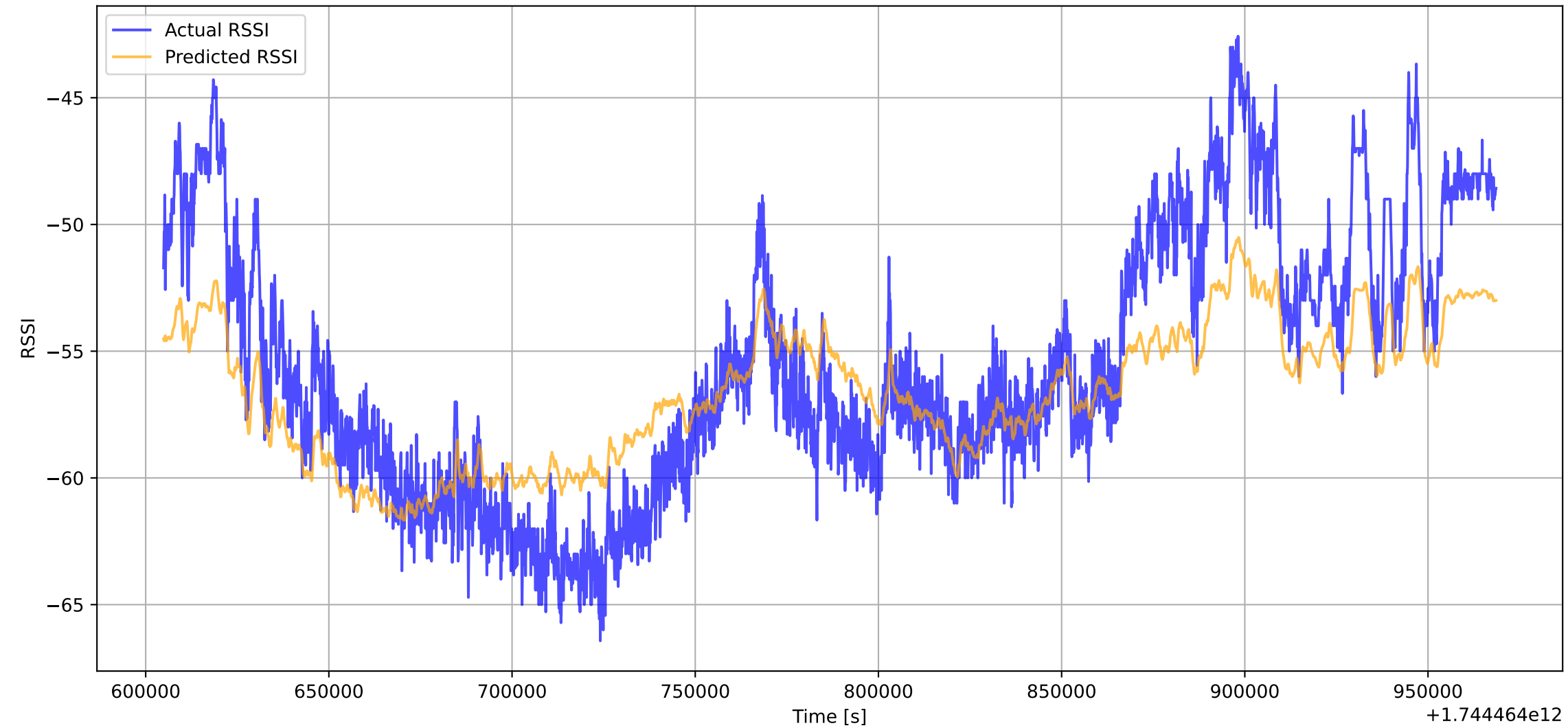
1744464597710
1744464597783
1744464597879
1744464597921
1744464598008

...

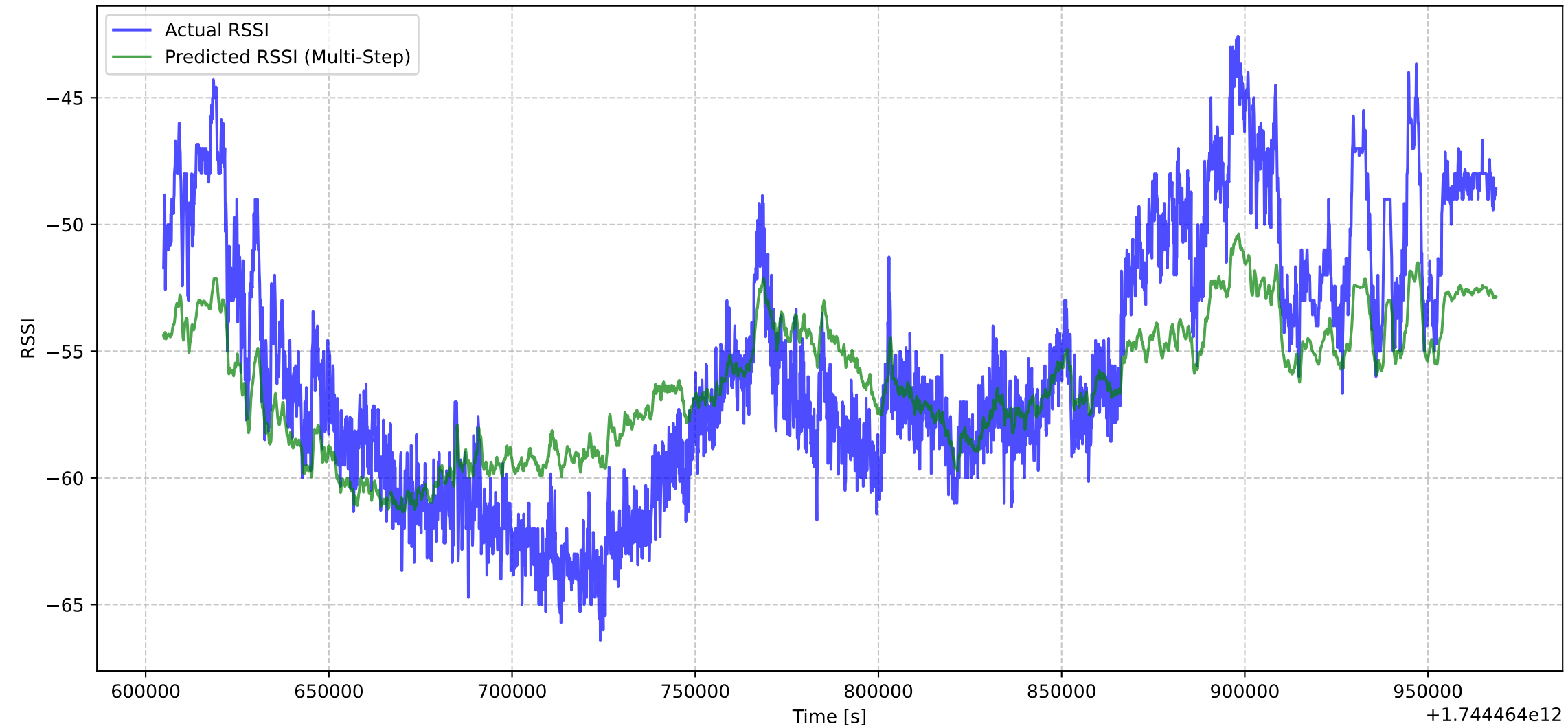
Last 5 timestamps AFTER:

1744464971041
1744464971124
1744464971209
1744464971281
1744464971366

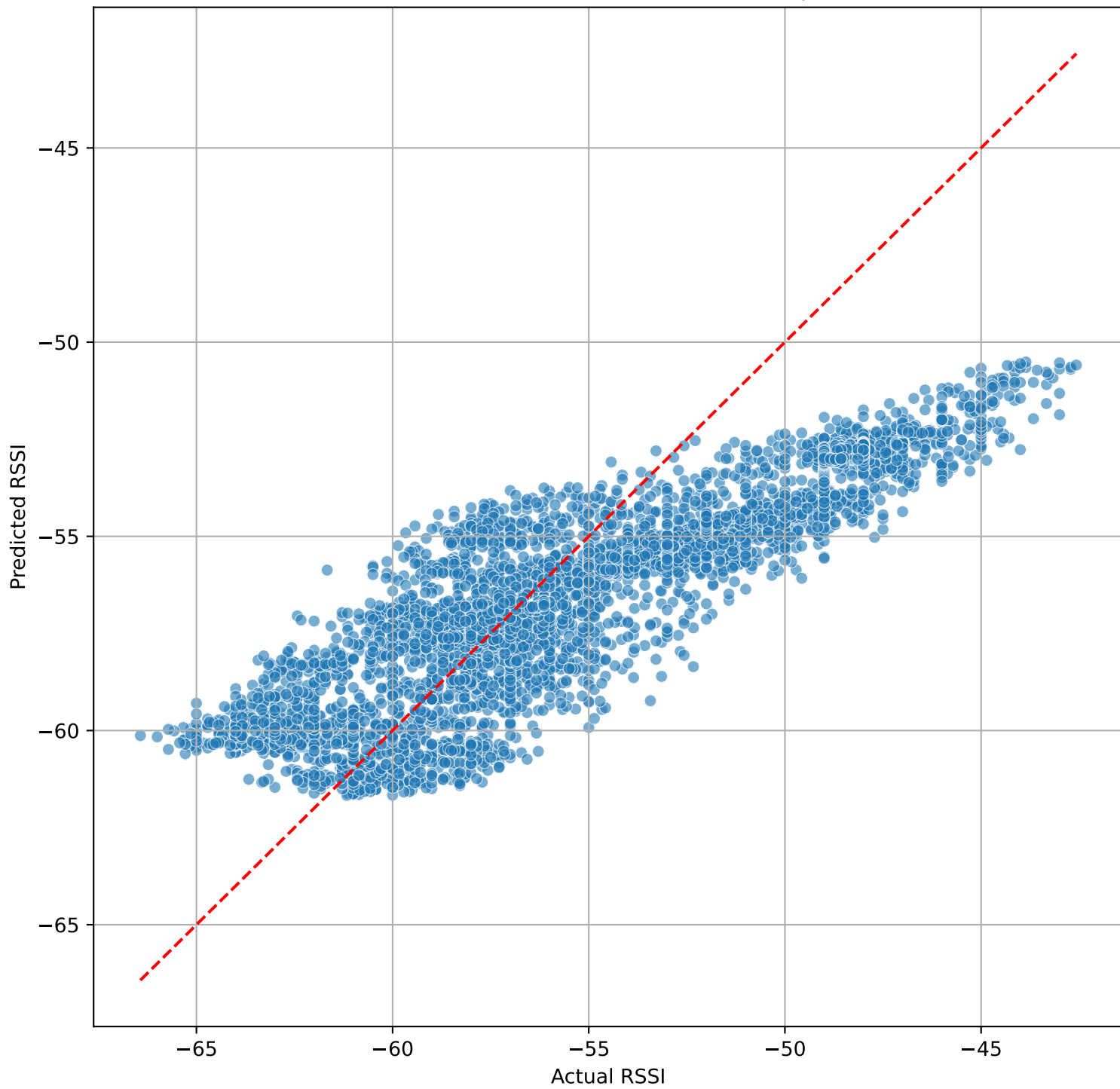
Actual vs Predicted RSSI (First Step)



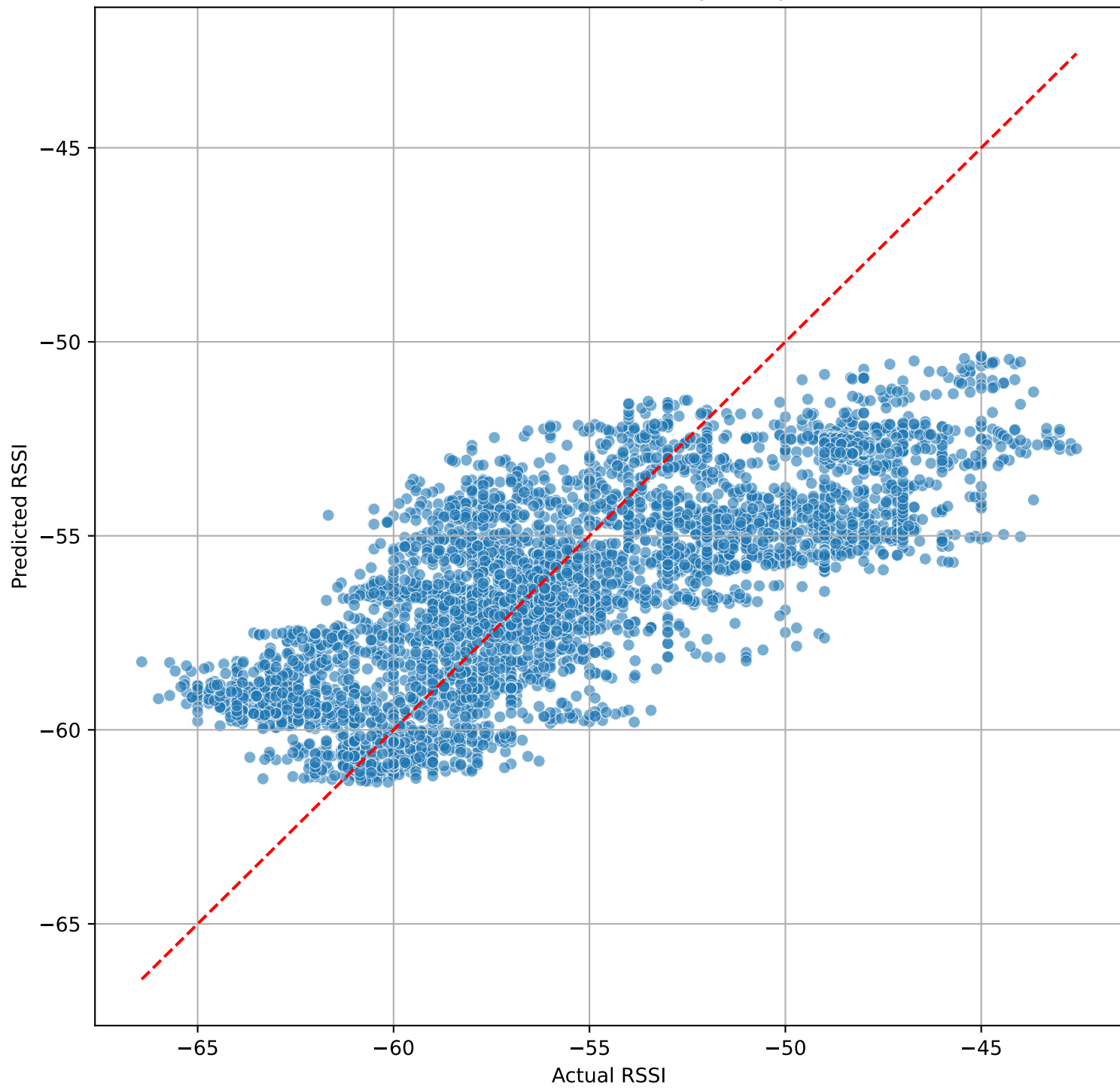
Actual vs Predicted RSSI (Multi-Step (step 36 $\approx$ 3.0 s))



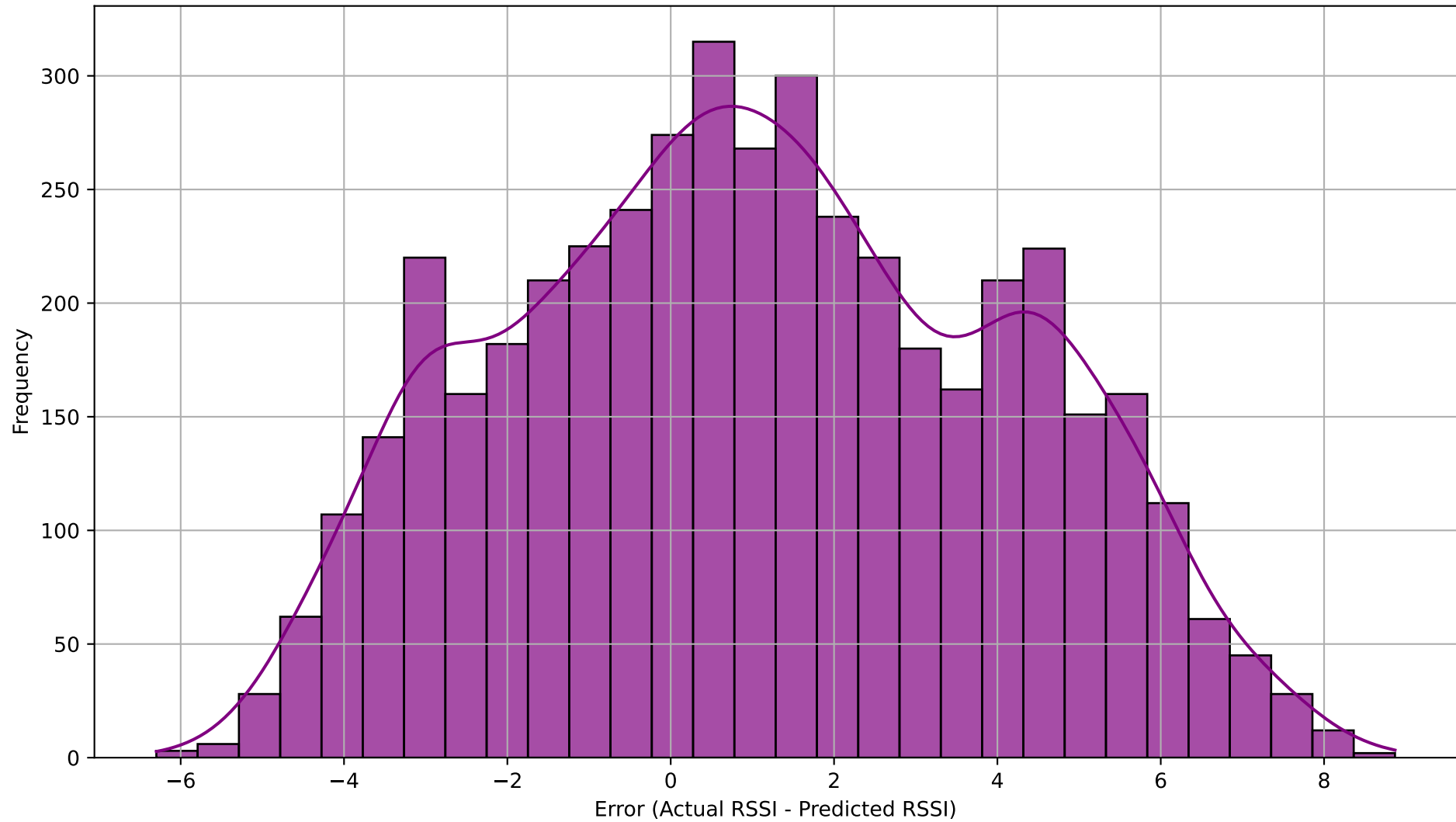
Actual vs Predicted RSSI (First Step)



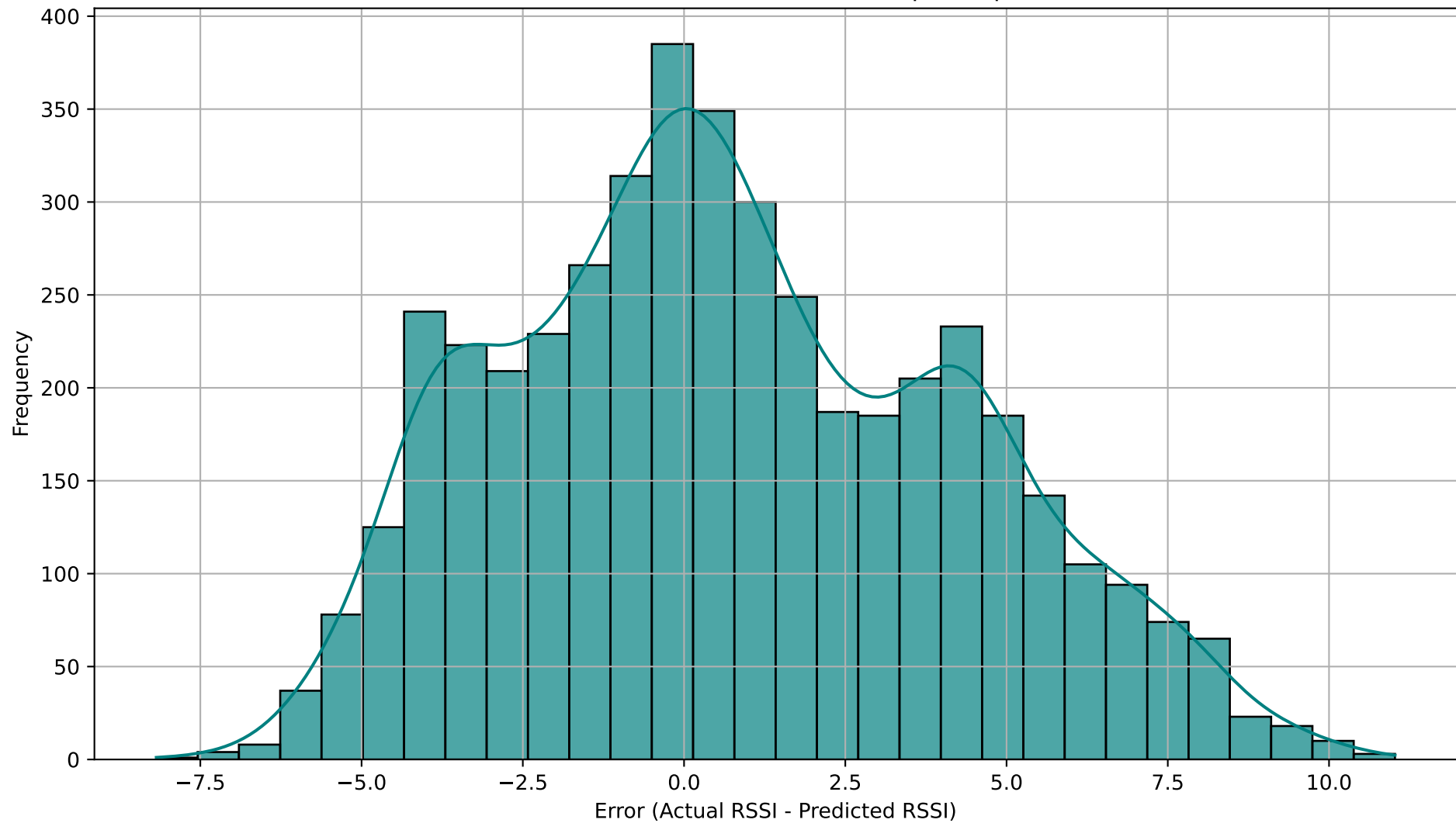
Actual vs Predicted RSSI (Last Step - step 36 $\approx$ 3.0 s)



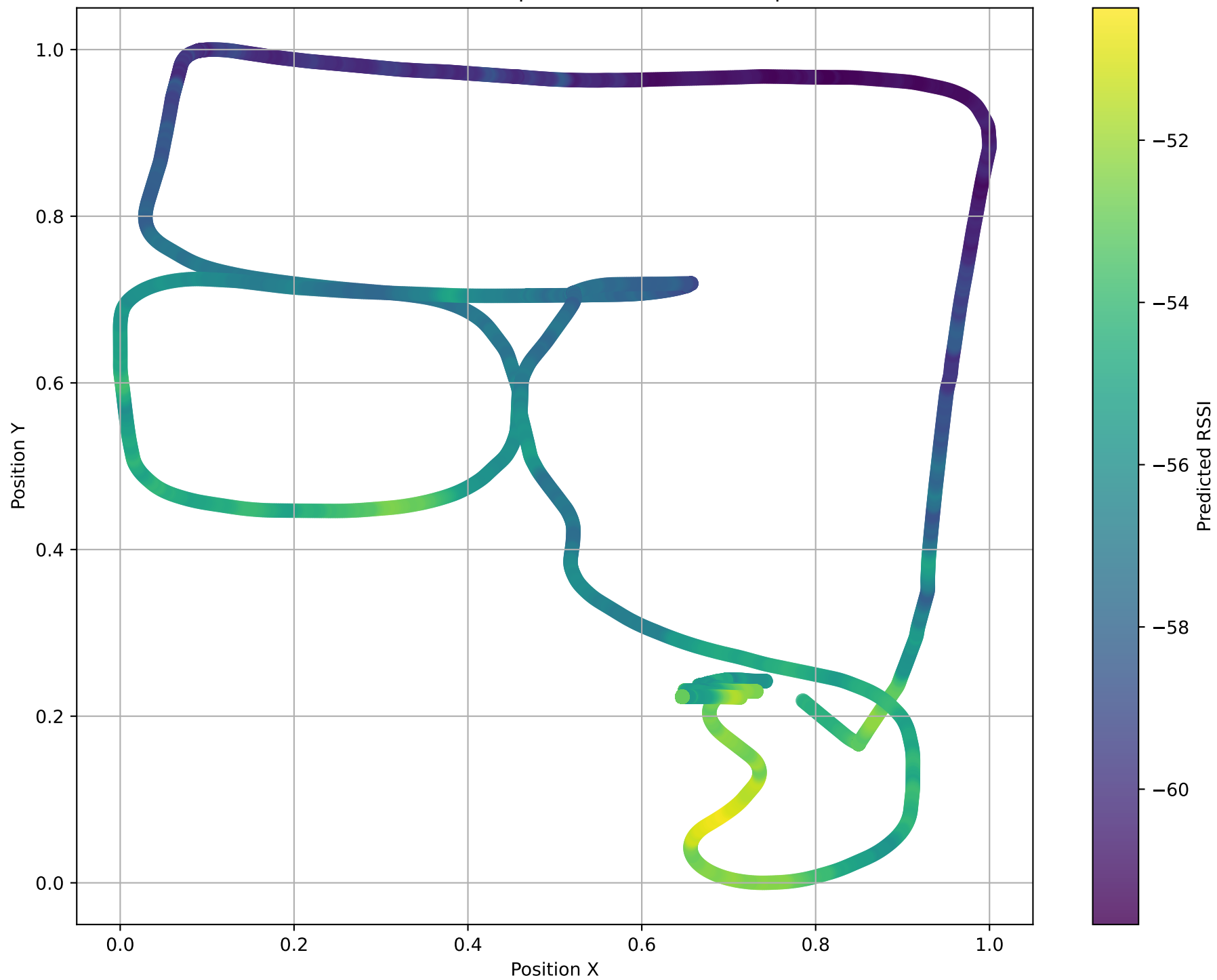
Distribution of Prediction Errors (First Step)



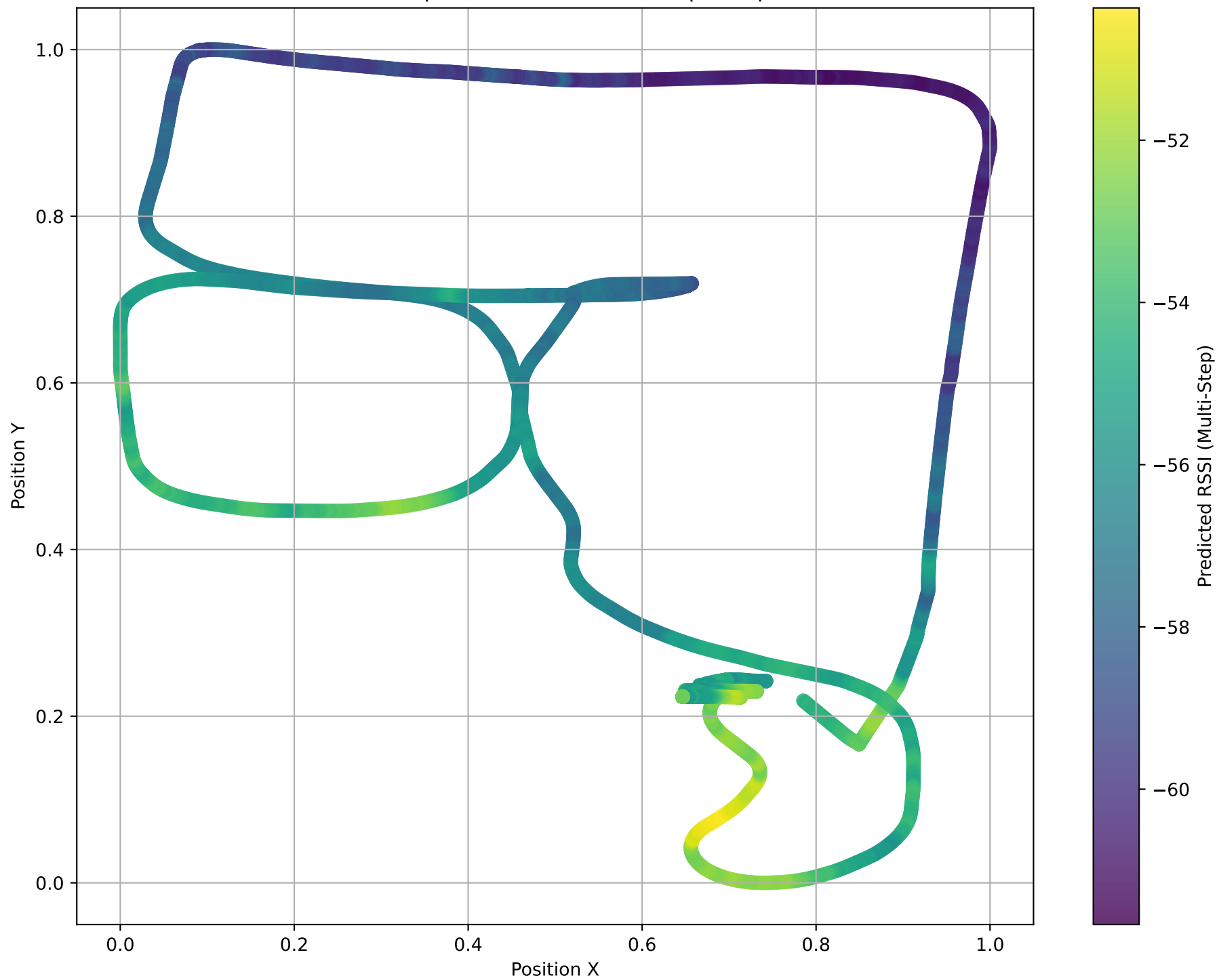
Distribution of Prediction Errors (Last Step - step 36 $\approx$ 3.0 s)



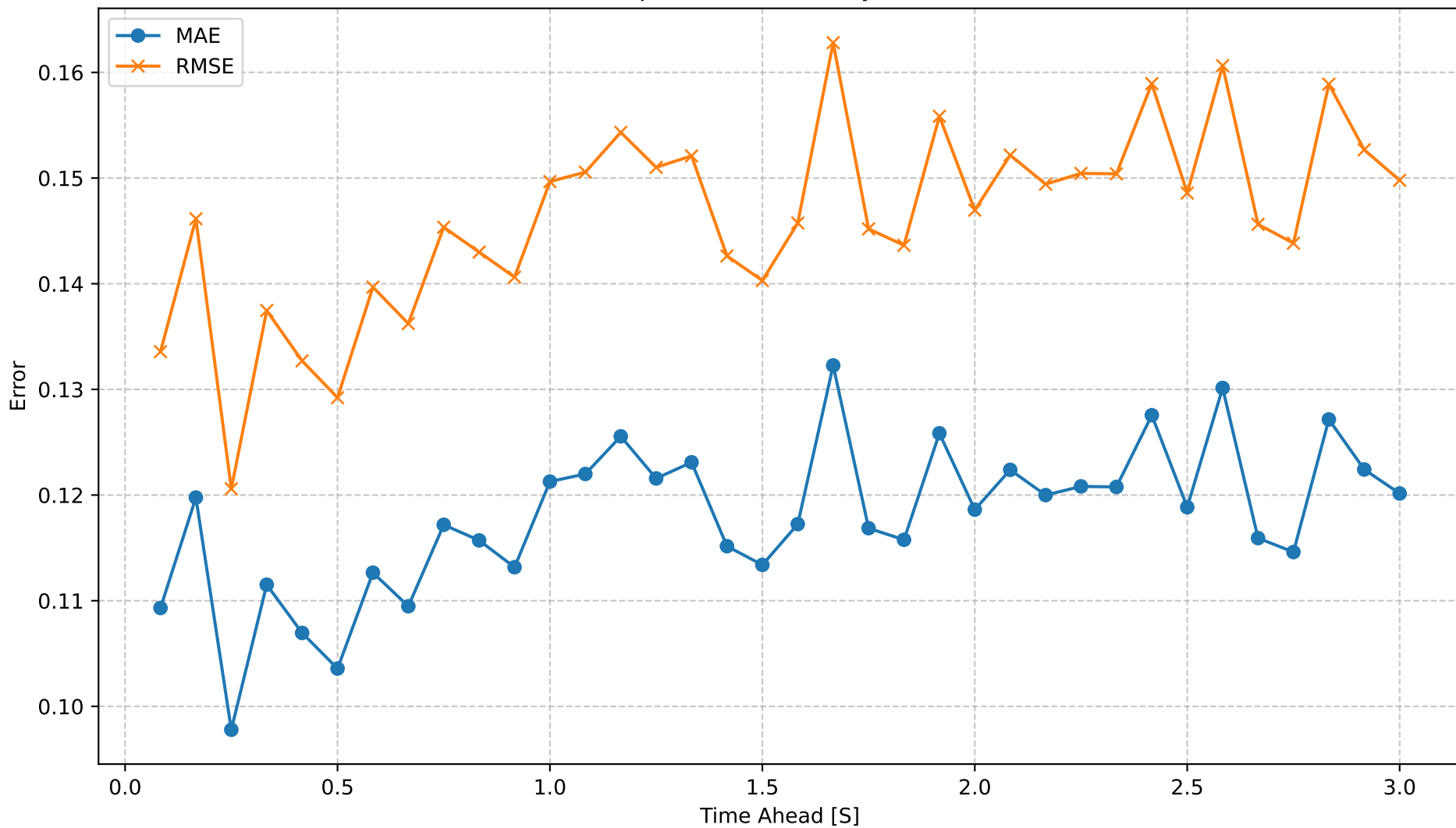
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 36 $\approx$ 3.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_11apr25_isolado.parquet |
Experiment: rssi_odom_dataset_12apr25_isolado.parquet
Modelo: LSTM
Forecast steps: 36

MAE / RMSE (dB)

- 1º passo : 0.11 / 0.13
- Mediana : 0.12 / 0.15
- 36º passo : 0.12 / 0.15

Crescimento do erro: 9.9% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 20 (0.13), 31 (0.13), 29 (0.13)

Recomendação:

O modelo é adequado para decisões de hand-over até o horizonte máximo configurado.